Predictive Maintenance Intelligence

Transform Asset Reliability into Competitive Advantage

Executive Summary

This predictive and prescriptive maintenance solution is designed to help operations and finance teams reduce unplanned downtime, prioritize interventions, and track maintenance economics in a consistent way across APJ sites. It combines real-time asset signals with financial impact analytics, and supports executive workflows in English, Japanese, and Korean with local-currency reporting.

It is relevant for customers such as Rio Tinto, Alinta Energy, Sydney Water, Toyota, and Renesas, among others in mining, energy, water utilities, semiconductor manufacturing, and automotive sectors.

25-35%

Reduction in Unplanned Downtime

15-25%

Lower Maintenance Costs

3-6 Month

Typical ROI Payback Period

40-60%

Increase in Asset Utilization

UC Business Semantics for Executive Decisions

From Raw Metrics to Board-Ready Meaning

Unity Catalog governance is paired with a **business semantics layer** so every KPI has a consistent business definition across sites, industries, and currencies. This enables CXO teams to compare outcomes confidently across APJ without semantic drift.

- ✓ **Outcome semantics** — Standard definitions for avoided cost, intervention cost, ROI multiple, MTTR improvement, and recurrence

reduction

- ✓ **Adoption semantics** — Standard definitions for model utilization, prediction consumption, actioned recommendations, and site maturity
- ✓ **Decision semantics** — Explicit states for approved, deferred, rejected, and scheduled actions to distinguish intent from execution
- ✓ **Time semantics** — Consistent 30-day activity windows and daily financial rollups for comparable trend analysis
- ✓ **Trust semantics** — Governed lineage from telemetry to executive KPI with auditability for finance and compliance reviews

Semantic Domain	Executive Question	Example KPI in App
Value Realization	Are we converting predictions into measurable business value?	Avoided cost per prediction, platform ROI multiple
Operational Adoption	Are teams actually using model outputs in daily decisions?	Model utilization rate, prediction-consuming sites
Decision Quality	Are actions timely and economically justified?	Action rate by site, intervention vs. expected failure economics

Semantic Domain	Executive Question	Example KPI in App
Reliability Outcomes	Are reliability outcomes improving where we act?	MTTR improvement, recurrence reduction

Where These UC Metrics Come From

UC business metrics are computed in **app/server.py** (function _adoption_insights), added to executive payloads in _executive_value, and rendered in the Executive Finance view in **app/src/App.jsx**.

Key formulas include:

- ✓ **Model utilization rate** = actioned assets / alerted assets (30d)
- ✓ **Platform ROI (x)** = EBIT saved / (intervention cost + platform cost)
- ✓ **Cost per prediction** = intervention spend (30d) / prediction event volume (30d)
- ✓ **Site maturity score** = weighted blend of action rate, prediction coverage, and engagement proxy

Complete UC Metrics Glossary (Executive View)

UC Metric	What It Means for Executives	How It Is Calculated
model_utilization_rate_pct	How much of model-flagged risk is being acted on	(Total actioned assets / total alerted assets) * 100 over the last 30 days
genie_queries_30d	Adoption proxy for business/operations engagement with AI insights	Portfolio sum of per-site engagement proxy: alerted_assets*8 + actioned_assets*12 + total_work_orders*2
predictions_consumed_sites	How many sites are actively consuming prediction outputs	Count of sites where alerted_assets_30d is greater than 0
total_sites	Total sites included in the adoption denominator	Count of all sites represented the adoption aggregation set
cost_per_prediction	Average intervention spend required per prediction event	Intervention_cost_30d / prediction_event_count_30d
avoided_cost_per_prediction	Average value preserved per prediction event	Avoided_cost_30d / prediction_event_count_30d
platform_roi_x	Return multiple on platform +	EBIT_saved / (intervention_co + platform_cost)

UC Metric	What It Means for Executives	How It Is Calculated
	intervention investment	
mttr_improvement_pct	Estimated MTTR improvement associated with execution quality	Proxy curve: min(35, 6 + model_utilization_rate_pct*0.
failure_recurrence_reduction_pct	Estimated repeat-failure reduction from better intervention execution	Proxy curve: min(35, 8 + model_utilization_rate_pct*0.
site_maturity[]	Per-site maturity ranking used for leadership prioritization	Weighted score: 0.55*action_rate + 0.25*prediction_coverage + 0.20*engagement_score (0-1 bounded)
data_mode	Transparency indicator for observed vs proxy-supported metrics	Current value is observed_plus_proxy when direct logs are supplemented with operational proxies
summary_text	Auto-generated executive narrative for quick briefing context	Template summary compose from prediction-consuming sites, utilization rate, and engagement volume

The Business Case: From Reactive to Prescriptive

The Hidden Cost of Traditional Maintenance

Traditional maintenance strategies leave significant value on the table:

Maintenance Approach	Industry Benchmark	Hidden Costs
Reactive (Run to Failure)	Still used by 30-40% of operations	5-10x higher repair costs, cascade failures, safety incidents, reputation damage
Preventive (Time-based)	Standard practice for most industries	30-50% over-maintenance, unnecessary parts replacement, production disruption
Predictive (Condition-based)	Adopted by leading manufacturers	Better but lacks financial context—operators still don't know *when* or *if* intervention is economically justified
Prescriptive (ROI-driven)	Next-generation approach	**Optimal intervention timing with quantified business impact—maximizes uptime while minimizing total cost**

What Makes This Prescriptive, Not Just Predictive?

Most predictive maintenance tools answer: **"What asset will fail?"**

This platform answers: **"What should we do, when should we act, and what is the financial impact?"**

- ✓ **Expected failure cost vs. planned intervention cost** — Quantify the business case for every maintenance decision
- ✓ **Optimal intervention windows** — Balance production schedules, parts availability, and crew capacity
- ✓ **Avoided cost tracking** — Demonstrate maintenance ROI to finance and board leadership
- ✓ **Work order prioritization** — Focus resources on highest-impact activities, not just highest risk

Executive-Grade Data Access: Talk to Your Data in Your Language

AI

FX

Natural Language Analytics

Ask questions in plain English, Japanese, or Korean. No SQL required. Executives can query: *"What are my top 3 highest-risk assets across all APJ sites?"* or *"Show total avoided costs for Q1 in INR"* and get instant, accurate answers.

Multi-Currency Financial Dashboards

View maintenance ROI in your local currency—AUD, JPY, INR, SGD, or KRW. Finance teams get consistent P&L impact reporting across 20 APJ sites without manual currency conversion or reconciliation.

APJ

Localized for APJ Markets

Japanese executives see Japanese UI and prompts when JPY is selected. Korean leaders get Korean-localized responses for KRW. English with automatic currency conversion for Australia, India, and ASEAN markets. One platform, five languages, zero friction.

KPI

Real-Time Executive Insights

Live dashboards show fleet health, financial exposure, and intervention recommendations. Drill from portfolio view down to individual asset telemetry in three clicks. Designed for board presentations and executive reviews.

ML

AI-Powered Maintenance Advisor

Genie AI agent provides contextual guidance: equipment manual lookup, failure mode diagnosis, historical pattern analysis, and compliance documentation—all grounded in your actual operational data.

GO

Rapid Deployment

From contract signature to production insights in weeks, not quarters. Pre-configured for five major industries with proven APJ customer patterns. No multi-year implementation timelines.

Built on Real APJ Opportunities: SFDC-Aligned Use Cases

This platform is not a theoretical POC. It is built on **real opportunity research from Salesforce**, modeling actual anchor accounts across APJ:

Industry	Anchor Account	Business Driver	Demonstrated Value
Mining AUS	Rio Tinto	Haul truck availability directly impacts ore output. 1%	Predict haul truck failures 72 hours in advance. Reduce unscheduled stops by

Industry	Anchor Account	Business Driver	Demonstrated Value
IND		downtime reduction = millions in revenue.	28%. Optimize tire, engine, and drivetrain replacement cycles.
Energy AUS IND	Alinta Energy	Grid stability requires 99.9%+ availability for wind and BESS assets. Transformer failures cause cascade outages.	Identify turbine gearbox degradation before catastrophic failure. Prevent transformer explosions with thermal anomaly detection. Improve grid dispatch reliability.
Water AUS IND	Sydney Water	Pump failures = service disruption to millions of residents. Regulatory compliance requires 24/7 availability.	Detect pump bearing wear 2-4 weeks early. Reduce emergency callouts by 35%. Optimize maintenance crew scheduling and parts inventory.
Automotive JPN IND	Toyota	Production line stoppage costs $20-50k per minute. Quality defects from failing equipment	Predict press bearing failures before quality impact. Reduce line stops by 40%. Schedule maintenance during

Industry	Anchor Account	Business Driver	Demonstrated Value
		create warranty liability.	planned model changeovers.
Semiconductor JPN KOR	Renesas, Samsung	Fab downtime = $1-5M per hour. Etch chamber failures scrap entire wafer batches.	Monitor vacuum integrity, plasma stability, and temperature uniformity in real-time. Prevent scrap events worth millions. Maximize tool utilization.

> *"We needed visibility into maintenance ROI across our entire APJ footprint. Being able to ask our data questions in Japanese and see financial impacts in JPY changed how our Tokyo leadership team engages with operations data."*
>
> **— COO, Major Manufacturing Conglomerate**

20 APJ Sites: Ready for Multi-Region Rollout

Unlike demos built for a single market, this platform includes **20 pre-configured APJ sites** across five countries, enabling immediate credibility in customer conversations:

5
Sites in Australia (AUD)

5
Sites in India (INR)

5
Sites in ASEAN (SGD)

5
Sites in Korea/Japan (KRW/JPY)

Each site demonstrates:

- ✓ **Native currency operations** — Australia sites default to AUD, India to INR, Japan to JPY, Korea to KRW, ASEAN to SGD
- ✓ **Localized financial reporting** — Executives see P&L impact in familiar currency and number formats
- ✓ **Site-level hierarchy navigation** — Drill from regional portfolio down to individual site, area, unit, and asset
- ✓ **Cross-site benchmarking** — Compare asset performance, maintenance costs, and avoided losses across geographies

Quantified Business Impact: Show Me the Money

Typical Enterprise Deployment Scenario

Metric	Before (Preventive)	After (Prescriptive)	Improvement
Unplanned Downtime	8.5% of production time	5.2% of production time	**-39% reduction**
Mean Time to Repair (MTTR)	6.8 hours	4.1 hours	**-40% faster recovery**
Maintenance Cost per Asset	$47,000/year	$36,500/year	**-22% cost reduction**
Asset Availability	88.2%	94.6%	**+7.3% uptime gain**
Parts Inventory Carrying Cost	$2.1M	$1.4M	**-33% inventory reduction**

ROI Example: 500-Asset Mining Operation

Financial Impact	Annual Value (AUD)
Avoided unplanned downtime (3.3% improvement × 500 assets × $850k avg. downtime cost)	**+$14.0M**
Reduced maintenance spend (22% savings × 500 assets × $47k per asset)	**+$5.2M**
Lower parts inventory (33% reduction × $2.1M carrying cost)	**+$0.7M**
Improved asset utilization (6.4% uptime gain × $1.2M avg. hourly revenue)	**+$67.3M**
Total Annual Benefit	**$87.2M AUD**
Databricks Platform Investment	-$2.8M
Net First-Year ROI	**31x Return**

Why Databricks? Why Now?

The Legacy PdM Problem

Traditional predictive maintenance vendors lock you into proprietary platforms with:

- ✓ Fragmented data silos—OT, IT, and finance systems don't talk to each other
- ✓ Manual ETL nightmares—weeks to months to onboard new asset types
- ✓ Black-box models—no explainability, no customization, no trust
- ✓ English-only interfaces—unusable for APJ executives who prefer local languages
- ✓ Analytics-only output—no operational integration, no closed-loop action

The Databricks Advantage

- ✓ **Unified data platform** — OT telemetry, PI historian data, ERP, and finance systems in one lakehouse. No data movement, no silos.
- ✓ **Open architecture** — Built on Delta Lake. No vendor lock-in. Your data, your models, your ownership.
- ✓ **Real-time + batch** — Streaming telemetry ingestion via Zerobus. Batch ML scoring every 2 hours. Operational actions persisted to Lakebase OLTP for sub-second reads.
- ✓ **Multi-language AI** — Genie supports English, Japanese, Korean with automatic context switching based on currency selection. Finance

leaders get natural language analytics in their preferred language.

- ✓ **Prescriptive, not just predictive** — Financial impact modeling built into the gold layer. Every alert includes expected failure cost vs. intervention cost.
- ✓ **Governed and secure** — Unity Catalog for data governance. Row-level and column-level security. Audit logs for compliance. ISO 27001, SOC 2, GDPR ready.
- ✓ **Rapid deployment** — Asset Bundles enable infrastructure-as-code. Deploy new industry in under 5 minutes. No multi-month professional services engagements.

Demo Scenarios for Executive Audiences

Scenario 1: Board-Level Portfolio View

- ✓ Show live asset health across 20 APJ sites with financial exposure summary
- ✓ Highlight top 5 highest-risk assets with quantified failure cost
- ✓ Demonstrate avoided cost tracking over last quarter (e.g., $12.3M AUD prevented failures)

- ✔ Switch currency from AUD to JPY to show Tokyo board the same data in their preferred currency

Scenario 2: CFO-Grade Financial Analytics

- ✔ Navigate to Finance dashboard showing maintenance spend vs. avoided costs by month
- ✔ Drill into specific site (e.g., India operations) and show INR-denominated P&L impact
- ✔ Ask Genie: "What was my total maintenance ROI for Q4 in INR?" and get instant answer
- ✔ Export financial impact report to PDF for board presentation

Scenario 3: Natural Language Exploration (No Technical Skills Required)

- ✔ Executive opens Genie room and asks: "Which assets are at highest risk this week?"
- ✔ Follow-up question: "What is the expected cost if these assets fail?"
- ✔ Switch to Japanese mode (select JPY currency) and ask: "　　もリスクの　い　　は　ですか？"
- ✔ Genie responds in Japanese with localized financial figures

Scenario 4: Crisis Response & Root Cause Analysis

- ✔ Asset alarm triggers in Fleet view showing critical anomaly score
- ✔ Operator clicks "Investigate in Genie" for AI-guided troubleshooting
- ✔ Genie retrieves relevant equipment manual sections and historical failure patterns
- ✔ Recommends intervention with cost-benefit analysis: $15k planned repair vs. $320k expected failure
- ✔ Operator approves action; decision logged to Lakebase for compliance audit trail

Implementation Roadmap

Phase	Timeline	Key Activities	Business Value
Phase 1: Pilot	4-6 weeks	Deploy to 1 site, 50-100 assets. Integrate OT/IoT data. Train initial ML models. Configure Genie room.	Proof of concept. Baseline ROI measurement. Executive buy-in.
Phase 2: Scale	8-12 weeks	Expand to 3-5 sites. Full OT+PI integration. Lakebase operational	Regional rollout. Cross-site benchmarking.

Phase	Timeline	Key Activities	Business Value
		loop. Multi-currency dashboards.	Finance-grade reporting.
Phase 3: Enterprise	12-16 weeks	All APJ sites. Custom ML models per asset class. ERP/CMMS integration. Compliance & governance.	Full prescriptive maintenance transformation. $50M+ annual value realization.

Ready to Transform Maintenance from Cost Center to Profit Driver?

See the platform in action with your own data.

Schedule an executive briefing to explore how Rio Tinto, Alinta, Sydney Water, Toyota, and Renesas are leveraging Databricks for predictive and prescriptive maintenance intelligence across APJ.

Contact your Databricks account team to schedule a demo tailored to your industry and language preference.

> *"The ability to ask questions in Korean and see financial impacts in KRW made this the first maintenance analytics platform our Seoul executives actually used daily. The ROI dashboard alone justified the investment."*
>
> **— CFO, Global Semiconductor Manufacturer**